EV Battery Course — Module 13: Battery Safety, Failure Analysis and Functional Safety

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Battery safety is layered: prevent faults, detect them early, contain any that escape, and protect people when containment is tested. This module covers how cells fail, the hazards a pack poses to drivers, technicians and responders, the safety-engineering methods used to manage them, and real recalls analysed from root cause to corrective action. It focuses on engineering analysis and prevention; it does not describe how to trigger failures.

1. Learning objectives

- Explain the initiation and propagation of thermal runaway.
- Describe internal and external shorts, overcharge, overdischarge, mechanical damage, separator failure and manufacturing defects as root causes.
- Describe gas generation, venting and battery fire characteristics.
- Explain electrical hazards, isolation, HV service safety and crash-related risks.
- Apply hazard analysis (HARA, FMEA, FTA) and ISO 26262 concepts.
- Outline safe storage, transport and emergency response.
- Analyse real failure cases through root cause, detection, containment and corrective action.

2. Fundamentals: energy in a small space

A charged cell holds two kinds of energy: electrical energy (what you can draw out) and chemical energy (the flammable electrolyte, and oxygen that can be released by the cathode). In normal use only the first comes out, slowly, under control. In a failure, both can come out in seconds as heat and gas.

3. How cells fail

Root cause	Mechanism	Main prevention
Internal short (manufacturing defect)	Metal particle, burr, folded separator or tab misalignment bridges anode and cathode	Clean manufacturing, inspection, self-discharge screening

Root cause	Mechanism	Main prevention
Internal short (lithium dendrite)	Plating from cold or fast charging grows through separator	BMS charge maps, thermal management
External short	Damaged wiring, water ingress, crash	Fuses, pyrofuses, insulation, sealing, crash structure
Overcharge	BMS or charger fault pushes voltage too high	Redundant voltage monitoring, charger limits, CID
Overdischarge	Copper dissolves; on recharge forms dendrites	BMS cut-off, low-SOC storage rules
Mechanical damage	Crush, penetration, deformation in crash or underbody strike	Pack structure, underbody shields, crash load paths
Thermal abuse	External fire, failed cooling	Thermal management, barriers
Separator failure	Shrinkage, melting or puncture	Ceramic-coated separators, shutdown separators

Thermal runaway sequence (temperatures approximate; vary by chemistry):

1. Local heating from a short or abuse.
2. SEI decomposes (~80–120 °C), exposing anode to electrolyte → more heat.
3. Separator melts (~130–170 °C) → larger internal short.
4. Cathode decomposes (NMC ~200 °C, LFP ~250–270 °C), releasing oxygen (much less for LFP).
5. Electrolyte burns or vents; cell temperature can reach several hundred °C.
6. Hot gases and ejecta can trigger neighbouring cells: **propagation**.

4. Diagram: layers of protection

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flowchart LR
  A["Prevent<br/>cell quality, BMS limits, cooling, structure"] -->
B["Detect<br/>voltage, temperature, pressure, gas sensors"] -->
C["Contain<br/>thermal barriers, spacing, venting"] --> D["Protect
people<br/>warning, isolation, egress time, rescue info"]
```

Each layer assumes the previous one can fail; this is the "Swiss cheese" or defence-in-depth model.

5. Gas, venting and fire characteristics

- **Vent gas:** hydrogen, carbon monoxide, carbon dioxide, hydrocarbons and electrolyte vapours, plus HF from fluorine compounds. Flammable and toxic. Volumes are large: on the order of litres per Ah of cell capacity for nickel-rich cells (LFP generally produces less).
- **Fire behaviour:** jet-like flames from vents, re-ignition hours or days later, and very high heat output; deep-seated in the pack where water struggles to reach.
- **Explosion risk:** vent gas accumulating in an enclosed space (garage, container, pack housing) can deflagrate.
- **LFP vs NMC:** LFP runaway is slower and cooler with less oxygen release, but it still vents flammable gas.

6. Electrical hazards and HV service safety

- **Shock:** above 60 V DC, contact can be dangerous. EV packs are 300–800 V.
- **Isolation:** the HV system floats relative to chassis. A single insulation fault is not immediately dangerous, which is why the isolation monitor warns before a second fault creates a shock path.
- **Arc flash:** opening connectors under load or shorting tools across terminals can create arcs.
- **Service safety:** trained and authorised technicians; insulated tools rated to 1,000 V; class 0 gloves; lock-out/tag-out of the service disconnect; verify absence of voltage with a rated meter; waiting time for capacitors to discharge; HV warning labels (orange cables).

7. Crash-related risks

- Side-pole and underbody impacts can deform the pack and short cells.
- Design responses: crash load paths around the pack, side sills and extruded frames, underbody plates, pyrofuses triggered by the airbag controller, contactors opening on crash signal, and post-crash isolation requirements (UNECE R94/R95/R137 and Indian equivalents require HV safety after crash tests).
- Responders need rescue sheets showing HV components and cut zones (ISO 17840).

8. Hazard analysis and safety engineering

Tool	Use in batteries
HARA (hazard analysis and risk assessment, ISO 26262)	Identifies vehicle-level hazards (e.g. fire while driving) and assigns ASIL based on severity, exposure and controllability
DFMEA / PFMEA	Design and process failure modes, causes and controls
FTA (fault tree analysis)	Top-down: which combinations of faults lead to runaway
FMEDA	Hardware failure rates and diagnostic coverage for the BMS
Safety goals and safe state	e.g. "Prevent cell overcharge" (ASIL C/D); safe state = contactors open
SOTIF (ISO 21448)	Hazards from intended functionality limits (e.g. SOC estimation errors)

9. Engineering equations

$$E_{cell} = V_{nom} \times C \qquad \Delta T = \frac{E}{m\,c_p}$$

Short-circuit current:

$$I_{sc} = \frac{V_{pack}}{R_{pack} + R_{fault}}$$

Body current with an isolation fault:

$$I_{body} \approx \frac{V}{R_{iso} + R_{body}}$$

10. Worked examples

Example 1: why runaway temperatures are so high. A 512 Wh LFP cell (2.93 kg): electrical energy = 512 × 3,600 = 1.84 MJ. If that energy alone stayed in the cell: ΔT = 1,843,200 ÷ (2.93 × 1,000) = **629 K**. Chemical reactions add more. Containing this is a heat-sink and barrier problem.

Example 2: short-circuit current. Pack 438 V, R_pack 40 mΩ, fault 5 mΩ: I = 438 ÷ 0.045 = **9,733 A**. Contactors cannot break this safely; the fuse or pyrofuse must.

Example 3: the meaning of 100 Ω/V. At 400 V the minimum isolation is 40 kΩ. If someone touches an HV conductor and the chassis, with body resistance about 1 kΩ: I ≈ 400 ÷ 41,000 = **9.8 mA**, roughly the threshold region of perception and let-go for DC exposure. That is why isolation requirements are set where they are, and why a second fault is dangerous.

11. Case studies

Case A — Chevrolet Bolt EV recall (2020–2021)

- **Problem:** fires in parked, fully charged Bolts.
- **Root cause:** GM and LG identified two rare manufacturing defects occurring together in the same cell: a torn anode tab and a folded separator, from LG plants.
- **Detection:** field incidents, data analysis, teardown.
- **Containment:** owners told to charge to 90%, then avoid going below about 70 miles of range, park outside; software diagnostics added.
- **Corrective action:** recall of about 140,000 vehicles and replacement of battery modules or packs; LG reimbursed GM a large part of the cost.
- **Lesson:** rare cell defects scale into very expensive recalls; traceability and cell-level process control are essential.

Case B — Indian e-scooter fires (2022)

- **Problem:** several high-profile scooter fires across brands within weeks, in hot weather.
- **Findings:** government-appointed expert investigations reported issues in cells and pack design, including inadequate BMS functions, poor thermal design and cell quality concerns (details varied by company).
- **Containment:** voluntary recalls of thousands of scooters by several manufacturers.
- **Corrective action:** AIS-156 and AIS-038 amendments added requirements for cell-level testing, BMS functions, thermal propagation, IP protection, and traceability; many makers moved to LFP and tightened supplier quality.
- **Lesson:** cost pressure without engineering depth creates safety risk; regulation followed field failures.

Case C — Hyundai Kona Electric (2020–2021)

- **Problem:** fires, some while charging.
- **Root cause:** linked by Korean regulators to possible folded anode tabs in LG cells from a particular plant; BMS software update was the first remedy.
- **Corrective action:** replacement of packs in about 82,000 vehicles globally.

- **Lesson:** software containment may not be enough when the root cause is physical.

12. Safe storage, transport and emergency response

- **Storage:** cells and packs stored at reduced SOC (often about 30% for transport), in fire-separated areas, with thermal monitoring and space between racks; damaged batteries quarantined in dedicated areas or containers.
- **Transport:** UN 38.3 testing; dangerous goods classes UN 3480 (cells/batteries), UN 3171 (vehicles); special rules for damaged or defective batteries.
- **Emergency response:** first responders use large volumes of water to cool packs, monitor for re-ignition for hours, use thermal cameras, keep vehicles away from buildings after an incident, and follow OEM rescue sheets. Do not attempt to open or cut into a damaged pack without training and equipment.

13. Manufacturing applications

- The biggest field-safety lever at a factory is **preventing internal short defects**: contamination control, burr control, alignment and self-discharge screening (Modules 11–12).
- HV safety in production: insulated stations, interlocks, trained operators, visual signs for live packs, and emergency response plans for a cell fire on the line.
- Safety documentation: DFMEA, PFMEA, control plans and functional-safety work products are audited by customers and regulators.

14. Common mistakes and misconceptions

- **"EVs catch fire more often than petrol cars."** Available data generally do not support this per vehicle, but EV fires are harder to extinguish and attract attention.
- **"LFP cannot catch fire."** It is much harder to ignite, but runaway and venting still occur.
- **"The BMS can stop a defect-driven internal short."** It can sometimes detect it early; it cannot remove the defect.
- **"HV is safe once the car is off."** The pack is always live internally; only the outside connections are disconnected.

15. Revision notes

- Root causes: internal shorts (defects, dendrites), external shorts, overcharge, overdischarge, mechanical and thermal abuse.
- Runaway: SEI → separator → cathode → electrolyte; propagation is the pack-level risk.

- Layers: prevent, detect, contain, protect.
- Isolation (100 Ω/V) limits fault current; second faults are dangerous.
- HARA/ASIL, FMEA and FTA structure safety work.

16. Glossary

Term	Meaning
HARA	Hazard analysis and risk assessment
ASIL	Automotive Safety Integrity Level
FTA	Fault tree analysis
SOTIF	Safety of the intended functionality (ISO 21448)
Pyrofuse	Pyrotechnic disconnect
LOTO	Lock-out/tag-out
UN 38.3	UN transport test for lithium batteries
Rescue sheet	Responder guide to a vehicle's HV system

17. Knowledge check

Conceptual

1. Why is a folded separator so dangerous?
2. Why can overdischarge lead to a later fire?
3. Why does the isolation monitor matter even though one fault is not immediately dangerous?
4. Why do battery fires re-ignite?
5. How does HARA decide an ASIL?

Numerical

1. A 3.6 V, 60 Ah NMC cell weighs 0.9 kg (c_p 1,000). Electrical energy and ΔT if absorbed?
2. A 350 V pack with R_pack 60 mΩ has a 10 mΩ external short. Current?
3. Minimum isolation resistance for an 800 V pack at 100 Ω/V?

Application

1. Propose a containment and investigation plan after two similar field fires on one model.
2. List safety measures for a battery storage warehouse at an Indian pack plant.

18. Answer key

Conceptual

1. It can expose anode and cathode directly, creating an internal short point that may only become active after cycling, making it hard to detect at the factory.
2. Copper from the anode current collector dissolves, then deposits as metal on recharging, forming conductive paths through the separator.
3. It detects the first insulation fault so it can be repaired before a second fault creates a path through a person or causes an arc.
4. Unreacted cells inside the pack can heat up later; water cannot reach deep inside; heat stored in the pack can trigger more cells.
5. By rating severity (S), exposure (E) and controllability (C) of each hazardous event; combinations map to QM or ASIL A–D.

Numerical

1. E = 3.6 × 60 = 216 Wh = 777.6 kJ; ΔT = 777,600 ÷ 900 = **864 K**.
2. 350 ÷ 0.070 = **5,000 A**.
3. 800 × 100 = **80 kΩ**.

Application

1. Containment: customer advisory (charge limit, park outdoors), stop-ship review, quarantine suspect lots. Investigation: retrieve BMS logs, compare VINs for common cell lots and production dates, teardown and CT of fire-adjacent cells, review supplier data, FTA; corrective action and verification; regulator notification.
2. Fire-separated storage rooms; reduced-SOC storage; spacing and rack limits; thermal and gas detection; sprinkler or water deluge suitable for lithium fires; quarantine area for damaged cells; trained response team; restricted access and HV signage; incident drills; FIFO and storage time limits.

19. Further learning

- Feng et al., "Thermal runaway mechanism of lithium ion battery for electric vehicles: A review," *Energy Storage Materials* 2018.
- ISO 26262, ISO 21448, ISO 6469-1/-3, ISO 17840 (rescue sheets).
- NHTSA and NTSB reports on EV fires and emergency response.
- NFPA guidance on EV fire response.
- MoRTH and ARAI notifications on AIS-156 / AIS-038 amendments.

Next: Module 14, recycling and the circular economy.